

Neural Network Classification of Atomic Defects

Input information, occlusion and generalization

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Research question and contribution

I investigated whether simulated electronic band-structure data could distinguish atomic-defect conditions in silicon. Beyond classification performance, I explored which information supported predictions, whether input sensitivity was consistent across models, and how performance changed with simulation conditions and input resolution.

My work covered dataset generation and preprocessing, neural-network implementation, training and evaluation, and exploratory analyses of model behavior. Electronic-structure inputs were generated using SIESTA, with structural variations prepared using Python and LAMMPS. The project used fully connected networks and CNNs, including PyTorch CNN implementations.

Inputs and prediction targets

The inputs were momentum-energy intensity maps of electronic structure. Separate classification tasks addressed vacancy concentration, Frenkel-pair concentration, and vacancy-interstitial separation. These were discrete condition classes rather than a general-purpose atom-by-atom defect counter.

Example electronic-structure input

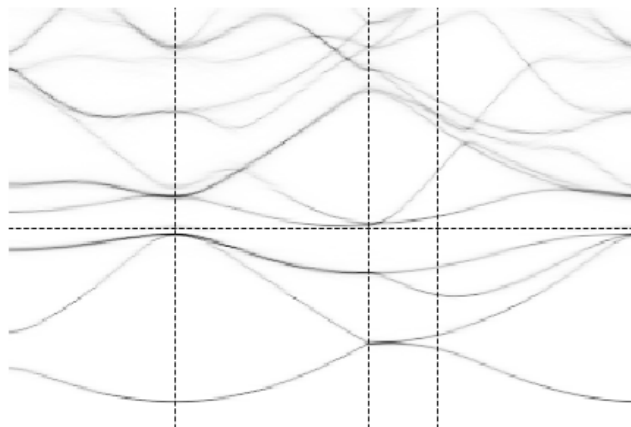


Figure 1. Example band-structure input from the historical meeting slides. The image encodes intensity across momentum and energy. Source: 0728, slide 16.

Experimental approach

I combined regional input occlusion, alternative input representations, and generalization tests. These experiments provided complementary checks on discriminative information, stability of model responses, and the conditions under which prediction succeeded or failed.

Regional occlusion and model dependence

How input regions affected predictions

I moved a zero-valued block across the input map and recomputed predictions. This produced a map of model responses to local input removal. The archived examples showed localized sensitivity in some cases, motivating the question of which energy-momentum regions supported the classification.

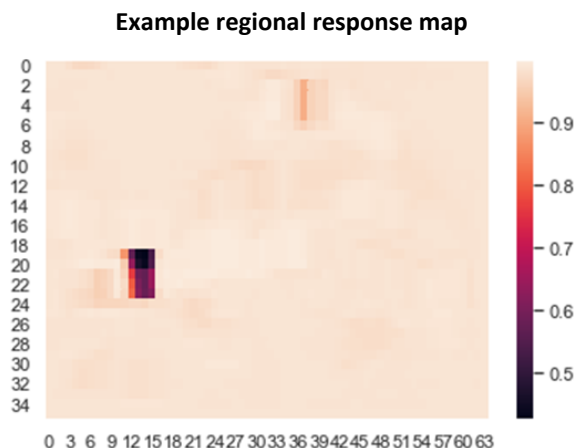


Figure 2. Archived occlusion map for a Frenkel-pair concentration example. Source: defense, slide 18. Its axes index block positions in the input.

Responses differed across models

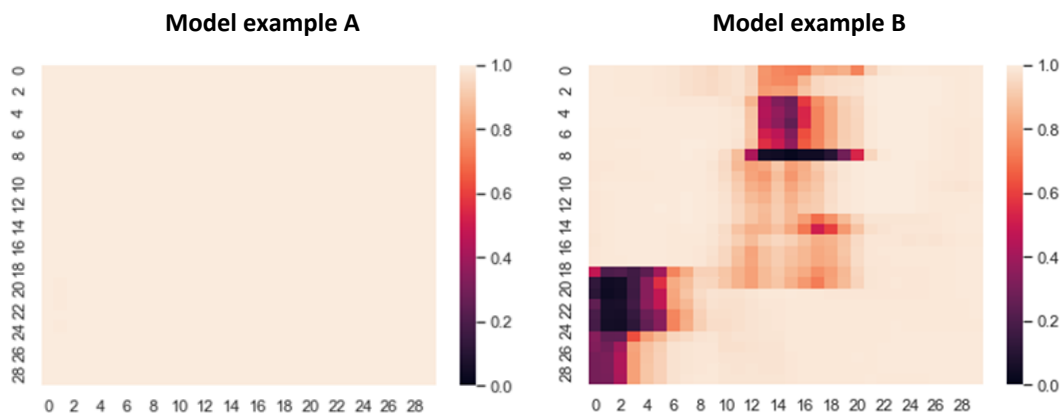


Figure 3. Two archived response maps for the same band-structure input, produced using different CNN models. Source: defense, slide 19.

The historical records explicitly note that different CNN models could yield different important regions. This exposed a limitation of treating one input-sensitivity map as a stable explanation. It raised questions about alternative predictive cues, distributed or redundant information, and sensitivity to the model and intervention.

These are input-level observations. They do not identify internal circuits or establish why the models differ. Zero masking also changes the input distribution, so its effect cannot automatically be interpreted as the removal of one isolated physical feature.

Measurement note: historical slides label these figures accuracy maps. Representative code stores class-output scores; incomplete figure-to-run metadata prevent a calibrated numerical interpretation. Here, they are called model-response maps.

Representations and generalization

Which information was sufficient for classification

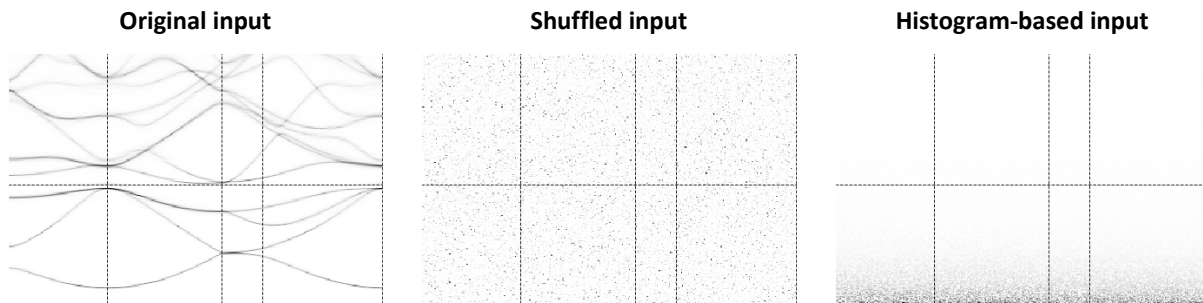


Figure 4. Archived examples of original, shuffled and histogram-based inputs. Source: 0728, slide 16. The last label follows the historical slides; it denotes an alternative input construction rather than a resolved explanation of the original CNN.

I compared histogram representations, shuffled inputs and thresholded band-structure maps to explore the contributions of intensity information and its arrangement across momentum and energy. The thresholded representation used a binary intensity cutoff; histogram classifiers were trained on reduced statistical descriptions.

The meeting records report successful histogram classification under some single-temperature conditions, with poorer classification for mixtures of temperatures. They also report that thresholded maps could support training. These observations motivated investigating which information remained predictive after the representation changed.

A key distinction: a classifier succeeding on a histogram representation shows that those statistics can support the task under the tested conditions. It does not demonstrate that the original CNN relied on those statistics, or that the original arrangement contributed nothing. This distinction was already stated in the historical discussion slides.

Generalization and unresolved questions

I tested changes in temperature, exchange-correlation settings, Gaussian blur and energy windows. The records show that transfer depended on the condition: some cross-functional tests performed well, while a model trained on one temperature performed substantially worse at a higher temperature. Blur augmentation improved the reported tolerance to reduced resolution.

Together, these experiments support exploratory interpretability and robustness analysis of neural classifiers. The contributions are the controlled comparisons, observed model dependence, and recognition of unresolved explanations. They leave open the internal mechanisms behind these behaviors and the reliability of transfer to measured experimental spectra.

Source record and scope

Figures are reproduced from original project slides. Text is grounded in the master thesis (pages 3-6), defense slides 18-19, and meeting slides 0728 (14-21), 0818 (5), and 0917 (13-18), together with archived code. This 2026 summary reorganizes historical work; it includes no new training runs or newly reproduced performance metrics.